

Probability and Statistics

Binomial Distribution

Dr. Yasir Ali (yali@ceme.nust.edu.pk)

DBS&H, CEME-NUST

May 11, 2022

Often, the observations generated by **different statistical experiments** have the **same general type of behavior**.

Consequently, discrete random variables associated with these experiments can be described by essentially the **same probability distribution** and therefore can be represented by **a single formula**.

In fact, one needs only a handful of important probability distributions to describe many of the discrete random variables encountered in practice.

Binomial Experiment

A **binomial experiment** is a probability experiment that satisfies the following four requirements:

- ① There must be a fixed number of trials.
- ② Each trial can have only two outcomes or outcomes that can be reduced to two outcomes. These outcomes can be considered as either success or failure.
- ③ The outcomes of each trial must be independent of one another.
- ④ The probability of a success must remain the same for each trial.

The outcomes of a binomial experiment and the corresponding probabilities of these outcomes are called a **binomial distribution**.

We are interested in the number of times an event A occurs in n independent trials. In each trial

the event A has the same probability $P(A) = p$

A will not occur with probability $q = 1 - p$

Let $X =$ Number of times the event A occurs in n trials. $0 \leq X \leq n$.

Now $X = x$ means that A occurs in x trials and in $n - x$ trials it does not occur. This may look as follows.

$$\underbrace{AAA \cdots A}_{x \text{ times}} \underbrace{BB \cdots B}_{n-x \text{ times}}$$

Here $B = A^c$ means A does not occur.

Binomial Distribution Function

$$P(x) = C_x^n (p)^x (q)^{n-x} = \frac{n!}{x!(n-x)!} p^x q^{n-x}$$

A coin is tossed 3 times. Find the probability of getting exactly two heads.

$HHH, \underbrace{HHT, HTH, THH}_{2\text{HEADS}}, HTT, THT, TTH, TTT$

A coin is tossed 3 times. Find the probability of getting exactly two heads.

$HHH, \underbrace{HHT, HTH, THH}_{2\text{HEADS}}, HTT, THT, TTH, TTT$

- 1 There are a fixed number of trials (three)

A coin is tossed 3 times. Find the probability of getting exactly two heads.

$HHH, \underbrace{HHT, HTH, THH}_{2\text{HEADS}}, HTT, THT, TTH, TTT$

- 1 There are a fixed number of trials (three)
- 2 There are only two outcomes for each trial, heads or tails

A coin is tossed 3 times. Find the probability of getting exactly two heads.

$HHH, \underbrace{HHT, HTH, THH}_{2\text{HEADS}}, HTT, THT, TTH, TTT$

- 1 There are a fixed number of trials (three)
- 2 There are only two outcomes for each trial, heads or tails
- 3 The outcomes are independent of one another (the outcome of one toss in no way affects the outcome of another toss)

A coin is tossed 3 times. Find the probability of getting exactly two heads.

$HHH, \underbrace{HHT, HTH, THH}_{2\text{HEADS}}, HTT, THT, TTH, TTT$

- 1 There are a fixed number of trials (three)
- 2 There are only two outcomes for each trial, heads or tails
- 3 The outcomes are independent of one another (the outcome of one toss in no way affects the outcome of another toss)
- 4 The probability of a success (heads) is in each case

A coin is tossed 3 times. Find the probability of getting exactly two heads.

$HHH, \underbrace{HHT, HTH, THH}_{2\text{HEADS}}, HTT, THT, TTH, TTT$

- ① There are a fixed number of trials (three)
- ② There are only two outcomes for each trial, heads or tails
- ③ The outcomes are independent of one another (the outcome of one toss in no way affects the outcome of another toss)
- ④ The probability of a success (heads) is in each case
 - Number of trials $n =$
 - X number of heads
 - $P(\text{Head}) = p$, $P(H^c) = P(\text{Tail}) = q$

$$P(x \text{ heads}) = C_x^n (p)^x (q)^{n-x} = C_x^n (1/2)^x (1/2)^{n-x}$$

$$P(x \text{ heads}) = C_x^n (p)^x (q)^{n-x} = C_x^n (1/2)^x (1/2)^{n-x}$$

x	0	1	2	3
f(x)	1/8	3/8	3/8	1/8

$$(p+q)^3 = \underbrace{C_3^3}_1 p^3 q^0 + \underbrace{C_2^3}_3 p^2 q^1 + \underbrace{C_1^3}_3 p q^2 + \underbrace{C_0^3}_1 p^0 q^3, \quad p = q = \frac{1}{2}$$

$$p + q = 1 \implies 1 = \frac{1}{8} + \frac{3}{8} + \frac{3}{8} + \frac{1}{8}$$

It is known that screws produced by a certain company will be defective with probability 0.01 , independently of each other. The company sells the screws in packages of 10 and offers a money-back guarantee that at most 1 of the 10 screws is defective. What proportion of packages sold must the company replace?

It is known that screws produced by a certain company will be defective with probability **0.01**, independently of each other. The company sells the screws in packages of 10 and offers a money-back guarantee that at most 1 of the 10 screws is defective. What proportion of packages sold must the company replace?

- Number of trials $n =$
- X number of Defective screws,
- $P(\text{Def}) = p =$, , $P(\text{ND}) = q =$

There are only two possible outcomes, Defective and Non-defective

Probability of Defective $p = 0.01$, Probability of Non-defective $q = 0.99$

Packages will not be replaced if there is 0 or 1 defective screws.

A package is replaced if $X \geq 2$. $P(X \geq 2) = ?$

$$P(X \geq 2) = ?, n = 10, p = 0.01, \& q = 0.99$$

$$P(X \geq 2) = ?, n = 10, p = 0.01, \& q = 0.99$$

$$p(x) = C_x^{10} (0.01)^x (0.99)^{10-x}$$

We know that $\sum_{x=0}^{n=10} P(x) = 1$

$$\begin{array}{c} \text{No def} \\ \underbrace{P(x=0)} \\ C_0^{10} (0.01)^0 (0.99)^{10} \end{array} + \begin{array}{c} \text{1 def} \\ \underbrace{P(x=1)} \\ C_1^{10} (0.01) (0.99)^9 \end{array} + P(x \geq 2) = 1$$
$$C_0^{10} (0.01)^0 (0.99)^{10} + C_1^{10} (0.01) (0.99)^9 + P(x \geq 2) = 1$$

$$P(x \geq 2) = 1 - (0.90438 + 0.09135) = \boxed{0.00427}$$

Example

A fair coin is tossed 6 times; call head as success. Find the probability that:

- (a) Exactly 2 heads occur.
- (b) At least 4 heads occur.
- (c) At least 1 head occur.

Example

A fair coin is tossed 6 times; call head as success. Find the probability that:

- (a) Exactly 2 heads occur.
- (b) At least 4 heads occur.
- (c) At least 1 head occur.

(a) $n = 6$, $x = 2$ and $n - x = 4$ with $p = q = 1/2$. The probability that exactly 2 heads occur is

$$P(X = 2) = C_2^6 p^2 q^4 = C_2^6 \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^4 = 15/64 = 0.23$$

Example

A fair coin is tossed 6 times; call head as success. Find the probability that:

(a) Exactly 2 heads occur.

(b) At least 4 heads occur.

(c) At least 1 head occur.

(a) $n = 6$, $x = 2$ and $n - x = 4$ with $p = q = 1/2$. The probability that exactly 2 heads occur is

$$P(X = 2) = C_2^6 p^2 q^4 = C_2^6 \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^4 = 15/64 = 0.23$$

(b) The probability that at least 4 heads occur is

$$P(4) + P(5) + P(6) = C_4^6 \left(\frac{1}{2}\right)^4 \left(\frac{1}{2}\right)^2 + C_5^6 \left(\frac{1}{2}\right)^5 \left(\frac{1}{2}\right) + C_6^6 \left(\frac{1}{2}\right)^6 \left(\frac{1}{2}\right)^0 = 0.34$$

(c) The probability of getting **No Head** is

$$q^6 = C_0^6 \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^6 = \left(\frac{1}{2}\right)^6 = 1/64. \text{ Thus the probability of one or more head is } 1 - q^6 = 1 - \frac{1}{64} = 0.94$$

Theorem

If X is a random variable with a binomial distribution with parameter n and p then

$$\mu = np, \quad \sigma^2 = npq$$

$$\begin{aligned} E[X] &= \sum_{x=0}^n x f(x) = \sum_{x=0}^n x \frac{n!}{x!(n-x)!} p^x q^{n-x} = \sum_{x=1}^n x \frac{n!}{x!(n-x)!} p^x q^{n-x} \\ &= \sum_{x=1}^n \frac{n!}{(x-1)!(n-x)!} p^x q^{n-x} = np \sum_{x=1}^n \frac{(n-1)!}{(x-1)!(n-x)!} p^{x-1} q^{n-x} \end{aligned}$$

Letting $y = x - 1$, $x = 1 \implies y = 0$ & $x = n \implies y = n - 1$

$$E[X] = np \sum_{y=0}^{n-1} \frac{(n-1)!}{(y)!(n-y-1)!} p^y q^{n-y-1} = np (p + q)^{n-1} = np$$

We know $E[X] = \mu = np$, $\sigma^2 = E[X^2] - (np)^2$ and $C_x^n = \frac{n}{x} C_{x-1}^{n-1}$

$$\begin{aligned}
 E[X^2] &= \sum_{x=0}^n x^2 C_x^n p^x q^{n-x} = \sum_{x=1}^n x n C_{x-1}^{n-1} p^x q^{n-x} = np \sum_{x=1}^n x C_{x-1}^{n-1} p^{x-1} q^{n-x} \\
 &\underbrace{=}_{y=x-1} np \sum_{y=0}^{n-1} (y+1) C_y^{n-1} p^y q^{n-y-1} = np \sum_{y=0}^{n-1} y C_y^{n-1} p^y q^{n-y-1} + np \sum_{y=0}^{n-1} C_y^{n-1} p^y q^{n-y-1} \\
 &= np \sum_{y=0}^{n-1} y C_y^{n-1} p^y q^{n-y-1} + np = np \sum_{y=0}^{n-1} (n-1) C_{y-1}^{n-2} p^y q^{n-y-1} + np \\
 &\underbrace{=}_{z=y-1} n(n-1)p^2 \sum_{z=0}^{n-2} C_z^{n-2} p^z q^{n-z-2} + np = n^2 p^2 - np^2 + np
 \end{aligned}$$

Thus $\sigma^2 = E[X^2] - (np)^2 = n^2 p^2 - np^2 + np - n^2 p^2 = np(1-p)$

Example

A coin is tossed 4 times. Find the mean, variance, and standard deviation of the number of heads that will be obtained.^a

$$^a n = 4, p = 1/2.$$

Example

A die is rolled 360 times. Find the mean, variance, and standard deviation of the number of 4s that will be rolled.^a

$$^a n = 360, p = 1/6.$$

In wheel of fortune a player bets on one of the numbers 1 through 6. Three dice are then rolled, and if the number bet by the player appears i times, $i = 1, 2, 3$, then the player wins i units; if the number bet by the player does not appear on any of the dice, then the player loses 1 unit. Is this game fair to the player?

letting X denote the player's winnings in the game, we have $X = -1, 1, 2, 3$

$$P\{X = -1\} = \binom{3}{0} \left(\frac{1}{6}\right)^0 \left(\frac{5}{6}\right)^3 = \frac{125}{216} \quad P\{X = 2\} = \binom{3}{2} \left(\frac{1}{6}\right)^2 \left(\frac{5}{6}\right)^1 = \frac{15}{216}$$

$$P\{X = 1\} = \binom{3}{1} \left(\frac{1}{6}\right)^1 \left(\frac{5}{6}\right)^2 = \frac{75}{216} \quad P\{X = 3\} = \binom{3}{3} \left(\frac{1}{6}\right)^3 \left(\frac{5}{6}\right)^0 = \frac{1}{216}$$

In order to determine whether or not this is a fair game for the player, let us calculate $E[X]$. From the preceding probabilities, we obtain

$$E[X] = \frac{-125 + 75 + 30 + 3}{216} = \frac{-17}{216}.$$

Hence, in the long run, the player will lose 17 units per every 216 games he plays.

Hypergeometric Distribution Sampling without Replacement.

When sampling is done without replacement, the binomial distribution does not give exact probabilities, since the trials are not independent. The smaller the size of the population, the less accurate the binomial probabilities will be.

If we wish to find the probability of observing 3 red cards in 5 draws from an ordinary deck of 52 playing cards, **the binomial distribution** does not apply unless each card is replaced and the deck reshuffled before the next draw is made.

To solve the problem of sampling without replacement, let us restate the problem. If 5 cards are drawn at random, we are interested in the probability of selecting 3 red cards from the 26 available in the deck and 2 black cards from the 26 available in the deck.

	Population	one type	other type
total	N=52	a=26	b=26
selected	n	x=3	n-x

$$\frac{C_3^{26} C_2^{26}}{C_5^{52}}$$

$$\frac{C_x^a C_{n-x}^b}{C_n^{a+b}}$$

Given a population with only two types of objects (females and males, defective and non-defective, successes and failures, etc.), such that there are a items of one kind and b items of another kind and $a + b$ equals the total population, the probability $P(X)$ of *selecting without replacement* a sample of size n with X items of type a and $n - X$ items of type b is

$$P(X) = \frac{C_X^a \times C_{n-X}^b}{C_n^{a+b}}$$

where

Items of one kind	$= a$
Items of other kind	$= b$
Total items	$= a + b$
Selection of sample size	$= n$
Selection of n must contain one kind	$= X$
Selection of n must contain other kind	$= n - X$

The probability distribution of the Hypergeometric random variable X , the number of successes in a random sample of size n selected from N items of which a are labeled success and $N - a$ labeled failure, is

$$h(x; N, n, a) = \frac{C_x^a C_{n-x}^{N-a}}{C_n^N}, \quad \max \{0, n - (N - a)\} \leq x \leq \min \{n, a\}.$$

Items of one kind	$= a$
Items of other kind	$= N - a$
Total items	$= a + b = N$
Selection of sample size	$= n$
Selection of n must contain one kind	$= X$
Selection of n must contain other kind	$= n - X$

$$P(x) = \frac{C_x^a \times C_{n-x}^b}{C_n^{a+b}}, \quad h(x; N, n, a) = \frac{C_x^a \times C_{n-x}^{N-a}}{C_n^N}$$

Example

Suppose a committee of 4 people is to be selected from 7 women and 5 men. What is the probability that the committee will consist of 3 women and 1 man?^a

^aIn this example, $a = ?$, $b = ?$, $n = ?$, $X = ?$, and $n - X = ?$.

Example

Suppose a committee of 4 people is to be selected from 7 women and 5 men. What is the probability that the committee will consist of 3 women and 1 man?^a

^aIn this example, $a = ?$, $b = ?$, $n = ?$, $X = ?$, and $n - X = ?$.

Women	Men	Total	(committee)	selection size(women)	selection size(men)
7	5	12	4	3	1
a	b	N	n	x	n-x

- 1 The number of ways a committee of 3 women and 1 man can be selected from 7 women and 5 men.
- 2 Total number of ways a committee of 4 people can be selected from 12 people.
- 3 $P(X) =$

Example

A recent study found that 2 out of every 10 houses in a neighborhood have no insurance. If 5 houses are selected from 10 houses, find the probability that exactly 1 will be uninsured.^a

^aIn this example, $a = 2$, $b = 8$, $n = 5$, $X = 1$, and $n - X = 4$.

Insured	Uninsured	Total
2	8	10
a	b	N

A particular part that is used as an injection device is sold in lots of 10. The producer deems a lot acceptable if no more than one defective is in the lot. A sampling plan involves random sampling and testing 3 of the parts out of 10. If none of the 3 is defective, the lot is accepted. Comment on the utility of this plan.

Let us assume that the lot is truly unacceptable (i.e., that 2 out of 10 parts are defective). The probability that the sampling plan finds the lot acceptable is

$$P(X = 0) = \frac{C_0^2 \cdot C_3^8}{C_3^{10}} = 0.467$$

Thus, if the lot is truly unacceptable, with 2 defective parts, this sampling plan will allow acceptance roughly 47% of the time. As a result, this plan should be considered faulty.

Lots of 40 components each are deemed unacceptable if they contain 3 or more defectives. The procedure for sampling a lot is to select 5 components at random and to reject the lot if a defective is found. What is the probability that exactly 1 defective is found in the sample if there are 3 defectives in the entire lot?

$$N = 40, n = 5, a = 3, b = 37, \text{ and } x = 1$$

A box contains 25 items, 10 of which are defective. A sample of two items will be taken, without replacement:

1. Find the probability and the distribution function.
2. What is the probability that
 - a. none of the items is defective?
 - b. at most one of the items is defective?

Let X denotes the number of defective items. $X = 0, 1, 2$

X	0	1	2
$f(X)$	0.35	0.5	0.15

Example

A purchaser of electrical components buys them in lots of size 10. It is his policy to inspect 3 components randomly from a lot and to accept the lot only if all 3 are non-defective. If 30 percent of the lots have 4 defective components and 70 percent have only 1, what proportion of lots does the purchaser reject?

Let A denote the event that the purchaser accepts a lot. Now A^c denote *the event that the purchaser reject a lot*. We know that $P(A^c) = 1 - P(A)$.

$$\begin{aligned} P(\text{Accepting a lot}) &= P(A|\text{lot has 1 defective}).P(\text{lot has 1 defective}) + \\ &\quad P(A|\text{lot has 4 defectives}).P(\text{lot has 4 defectives}) \\ &= \frac{C_0^4 C_3^6}{C_3^{10}} \left(\frac{3}{10} \right) + \frac{C_0^1 C_3^9}{C_3^{10}} \left(\frac{7}{10} \right) = \frac{54}{100} \end{aligned}$$

Hence, 46 percent of the lots are rejected.

A manufacturer of automobile tires reports that among a shipment of 5000 sent to a local distributor, 1000 are slightly blemished. If one purchases 10 of these tires at random from the distributor, what is the probability that exactly 3 are blemished?

Hypergeometric

$$P(X = 3) = \frac{C_3^{1000} \cdot C_7^{4000}}{C_{10}^{5000}} = 0.201477715$$

Binomial¹

$$P(X = 3) = C_3^{10} \left(\frac{1}{5}\right)^3 \cdot \left(\frac{4}{5}\right)^7 = 0.201326592$$

¹As a rule of thumb, if the population size is more than 20 times the sample size ($N > 20n$), then we may use binomial probabilities in place of Hypergeometric probabilities.

A discrete probability distribution that is useful when n is large and p is small and when the independent variables occur over a period of time is called the **Poisson distribution**.

In addition to being used for the stated conditions (i.e., n is large, p is small, and the variables occur over a period of time), the Poisson distribution can be used when a density of items is distributed over a given area or volume, such as the number of plants growing per acre or the number of defects in a given length of videotape.

Formula for Poisson Distribution Function

The probability of X occurrences in an interval of time, volume, area, etc., for a variable where λ (Greek letter lambda) is the mean number of occurrences per unit (time, volume, area, etc.) is

$$f(x) = \frac{\lambda^x}{x!} e^{-\lambda} \quad \text{where } X = 0, 1, 2, \dots$$

The letter e is a constant approximately equal to 2.7183
Round the answers to four decimal places.

Example

If there are 200 typographical errors randomly distributed in a 500-page manuscript, find the probability that a given page contains exactly 3 errors.

First, find the mean number λ of errors. Since there are 200 errors distributed over 500 pages, each page has an average of $\lambda = 200/500 = 0.4$. Since $X = 3$, substituting into the formula yields

$$f(3) = \frac{0.4^3}{3!} e^{-0.4} = 0.0072$$

Thus, there is less than a 1% chance that any given page will contain exactly 3 errors.

Example

If approximately 2% of the people in a room of 200 people are left-handed, find the probability that exactly 5 people there are left-handed.^a

^aThe Poisson distribution can also be used to approximate the binomial distribution when the expected value $\lambda = n.p$ is less than 5.

Example

If approximately 2% of the people in a room of 200 people are left-handed, find the probability that exactly 5 people there are left-handed.^a

^aThe Poisson distribution can also be used to approximate the binomial distribution when the expected value $\lambda = n.p$ is less than 5.

Poisson $\lambda = n.p = 200.(0.02) = 4$. Hence

$$f(x) = \frac{\lambda^x}{x!} e^{-\lambda} = \frac{4^5}{5!} e^{-4} = 0.1563.$$

Binomial $C_5^{200} (0.02)^5 (0.98)^{195} = 0.1579$

Example

If approximately 2% of the people in a room of 200 people are left-handed, find the probability that exactly 5 people there are left-handed.^a

^aThe Poisson distribution can also be used to approximate the binomial distribution when the expected value $\lambda = n.p$ is less than 5.

Poisson $\lambda = n.p = 200.(0.02) = 4$. Hence

$$f(x) = \frac{\lambda^x}{x!} e^{-\lambda} = \frac{4^5}{5!} e^{-4} = 0.1563.$$

Binomial $C_5^{200}(0.02)^5(0.98)^{195} = 0.1579$

Example

During a laboratory experiment, the average number of radioactive particles passing through a counter in 1 millisecond is 4. What is the probability that 6 particles enter the counter in a given millisecond?^a

^a $\lambda = 4, x = 6$.

A sales firm receives, on average, 3 calls per hour on its toll-free number. For any given hour, find the probability that it will receive the following.

a. At most 3 calls, b. At least 3 calls c. 5 or more calls

- (a) “At most 3 call” means 0, 1, 2, or 3 calls. Hence, $P(\text{At most 3 call}) = P(X=0) + P(X=1) + P(X=2) + P(X=3)$
- (b) “At least 3 calls” means 3 or more calls. It is easier to find the probability of 0, 1, and 2 calls and then subtract this answer from 1 to get the probability of at least 3 calls.
- (c) For the probability of 5 or more calls, it is easier to find the probability of getting 0, 1, 2, 3, or 4 calls and subtract this answer from 1.

Example

If the probability of producing a defective screw is $p = 0.01$, what is the probability that a lot of 100 screws will contain more than 2 defectives?

The complementary event is A^c *Not more than 2 defectives*. For its probability we get, from the binomial distribution ($n = 100, p = 0.01, q = 0.99$)

$$P(A^c) = C_0^{100}(0.01)^0(0.99)^{100} + C_1^{100}(0.01)^1(0.99)^{99} \\ + C_2^{100}(0.01)^2(0.99)^{98} = 0.9206$$

Thus $P(A) = 1 - 0.9206 = 0.0794$

Now find $P(A^c)$ using Poisson Distribution using $\lambda = 100(0.01) = 1$ in $f(x) = \frac{\lambda^x e^{-\lambda}}{x!}$ for $x = 0, 1, 2$

$$f(0) + f(1) + f(2) = e^{-1}(1 + 1 + \frac{1}{2!})$$

Both the Expected Value and the Variance of a Poisson Random Variable would equal its parameter λ

$$E[X] = \sigma^2 = \lambda$$

$$\begin{aligned} E[X] &= \sum_x x \cdot f(x) = \sum_{x=0}^{\infty} x \cdot \left(\frac{\lambda^x e^{-\lambda}}{x!} \right) \\ &= \sum_{x=1}^{\infty} x \cdot \left(\frac{\lambda^x e^{-\lambda}}{x!} \right) \\ &= e^{-\lambda} \sum_{x=1}^{\infty} \left(\frac{x \cdot \lambda^x}{x!} \right) \\ &= e^{-\lambda} \sum_{x=1}^{\infty} \left(\lambda \frac{x \cdot \lambda^{(x-1)}}{x!} \right) \\ &= \lambda \cdot e^{-\lambda} \sum_{x=1}^{\infty} \left(\frac{\lambda^{(x-1)}}{(x-1)!} \right) \\ &= \lambda \cdot e^{-\lambda} \sum_{i=0}^{\infty} \left(\frac{\lambda^i}{i!} \right) \quad \text{letting } x - 1 = i \end{aligned}$$

We know that $\sum_{i=0}^{\infty} \frac{\lambda^i}{i!} = e^{\lambda}$. Thus

$$E[X] = \lambda e^{-\lambda} \cdot e^{\lambda} \Rightarrow E[X] = \lambda$$

For Poisson Distribution $f(x) = \frac{\lambda^x}{x!} e^{-\lambda}$, mean is λ . We can calculate variance as follows

$$\sigma^2 = E[X^2] - \mu^2 = E[X(X-1)] + E[X] - \mu^2$$

$$E[X(X-1)] = \sum_{x=0}^{\infty} x(x-1) \frac{\lambda^x}{x!} e^{-\lambda} = \lambda^2 \sum_{x=2}^{\infty} \frac{\lambda^{x-2}}{(x-2)!} e^{-\lambda}$$

Letting $x-2 = y$ gives

$$E[X(X-1)] = \lambda^2 \sum_{y=0}^{\infty} \frac{\lambda^y}{y!} e^{-\lambda} = \lambda^2.$$

$$\text{As } \sum_{y=0}^{\infty} \frac{\lambda^y}{y!} = e^{\lambda}.$$

Thus

$$\sigma^2 = E[X(X-1)] + E[X] - \mu^2 = \lambda^2 + \lambda - \lambda^2 = \lambda$$

Binomial Distribution approaches to Poisson Distribution

We know that $f_b(x) = \frac{n!}{x!(n-x)!} p^x q^{n-x}$ here $p = n.\lambda$. As $\lim_{n \rightarrow \infty}$, $f_b(x)$ takes the following form

$$\lim_{n \rightarrow \infty} \frac{n!}{x!(n-x)!} \left(\frac{\lambda}{n}\right)^x \left(1 - \frac{\lambda}{n}\right)^{n-x}$$
$$= \lim_{n \rightarrow \infty} \frac{\overbrace{n(n-1) \cdots (n-x+1)}^{x \text{ terms}}}{x!} \frac{\lambda^x}{n^x} \left(1 - \frac{\lambda}{n}\right)^n \left(1 - \frac{\lambda}{n}\right)^{-x}$$

$$\text{Now } \lim_{n \rightarrow \infty} \frac{\overbrace{n(n-1) \cdots (n-x+1)}^{x \text{ terms}}}{n^x} = \lim_{n \rightarrow \infty} \left[(1) \left(1 - \frac{1}{n}\right) \cdots \left(1 - \frac{x-1}{n}\right) \right] = 1$$

$$\lim_{n \rightarrow \infty} \left(1 - \frac{\lambda}{n}\right)^{-x} = 1$$

$$\lim_{n \rightarrow \infty} \left(1 - \frac{\lambda}{n}\right)^n = e^{-\lambda} \text{ as } \lim_{n \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x = e.$$

Thus

$$\lim_{n \rightarrow \infty} f_b(x) = \frac{\lambda^x}{x!} e^{-\lambda}$$

A biased die with six faces is rolled, the random variable X denote the number on the upper face. The probability distribution is shown here

X	1	2	3	4	5	6
$P(X=x)$	a	a	b	b	b	0.4

Given $E[X] = 4.2$ find the values of a and b . Find $Var(5 - 3X)$.

Patients arrive at hospital accident and emergency department at random rate of 10 per hour

- (a) Find the probability that during any 90 minutes period, the patients arrive at hospital accident and emergency department are
- 1 Exactly 12
 - 2 at least 15
- (b) A patient arrive at 11:30. find the probability that next patient arrives before 11:45.

(a)

$$\lambda = 10/\text{hour}, \lambda_{90} = 15/90 \text{ min}, , \quad P(x) = \frac{\lambda_{90}^x e^{-\lambda_{90}}}{x!}$$

$$P(x \leq 15) + P(x > 15) = 1$$

- (b) Let $\hat{\lambda}$ be the number of patient per 15 min then $\hat{\lambda} = 2.5$

$$P(x \geq 1) = 1 - P(0)$$